

AI · ESG · RAG SYSTEM

ESG RAG Chatbot

Production-grade LLM System

A production-grade RAG chatbot designed for the ESRS corpus
Grounded answers · Cited sources · Built for real-world use

Rudy Nzau

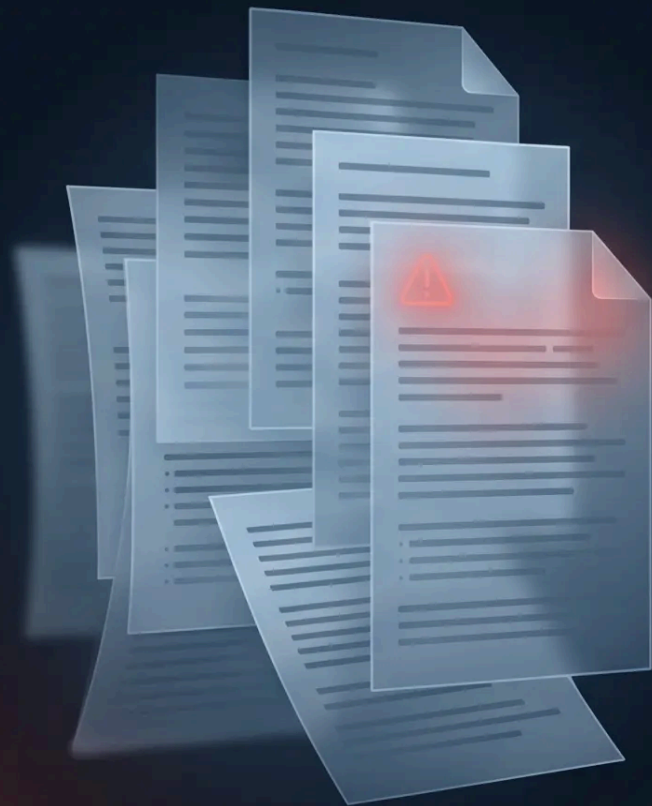
Data Scientist · AI Engineer · CentraleSupélec

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THE CHALLENGE

The Problem: Challenges for ESG Analysts



Hours Lost to Manual Search

Analysts spend days hunting through thousands of regulatory pages



CSRD · SFDR · ESRS — Thousands of Pages

No unified way to navigate across overlapping standards



No Natural Language Query

Cannot ask a question — must know exactly where to look



Risk of Missing Critical Disclosures

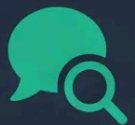
Missed requirements carry regulatory and reputational consequences

— *How do you make a 500-page regulation instantly queryable?*

SOLUTION

The Answer

A production-grade RAG chatbot that makes 500-page ESG regulations instantly queryable — with cited, grounded answers.



Natural Language Queries

Ask questions in plain English, no boolean syntax



Source-Grounded Responses

Every answer tied directly to ESRS corpus content



Explicit Citations

Responses include [1], [2], [3] source references



Production-Ready

Deployed on GCP Cloud Run — not just a demo

Technical Architecture

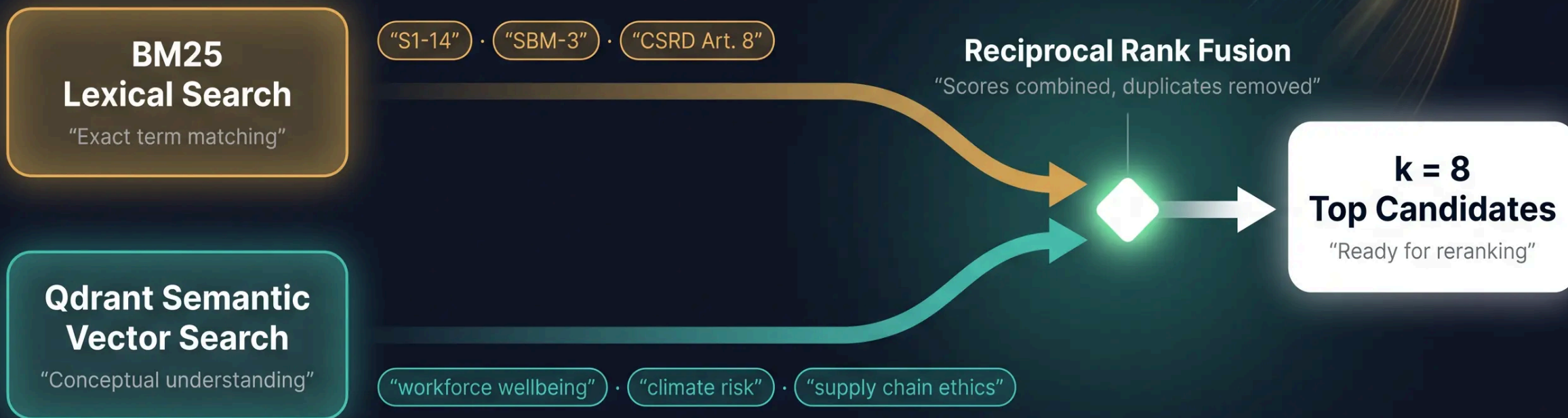
From raw PDF to grounded answer — every step in the chain



Hybrid Retrieval · CrossEncoder Reranking · Production Deployment on GCP

Hybrid Retrieval Strategy

Why one method is never enough for regulatory text



💡 "Semantic search alone misses **S1-14**' - BM25 alone misses the concept behind it. Hybrid captures both."

CrossEncoder Reranking

Scoring every (query, chunk) pair to keep only what matters

THE CHALLENGE

Retrieval returns 8 chunks — not all equally relevant to the query

HOW IT WORKS

CrossEncoder scores each (query, chunk) pair individually → Selects the top 4 by relevance score

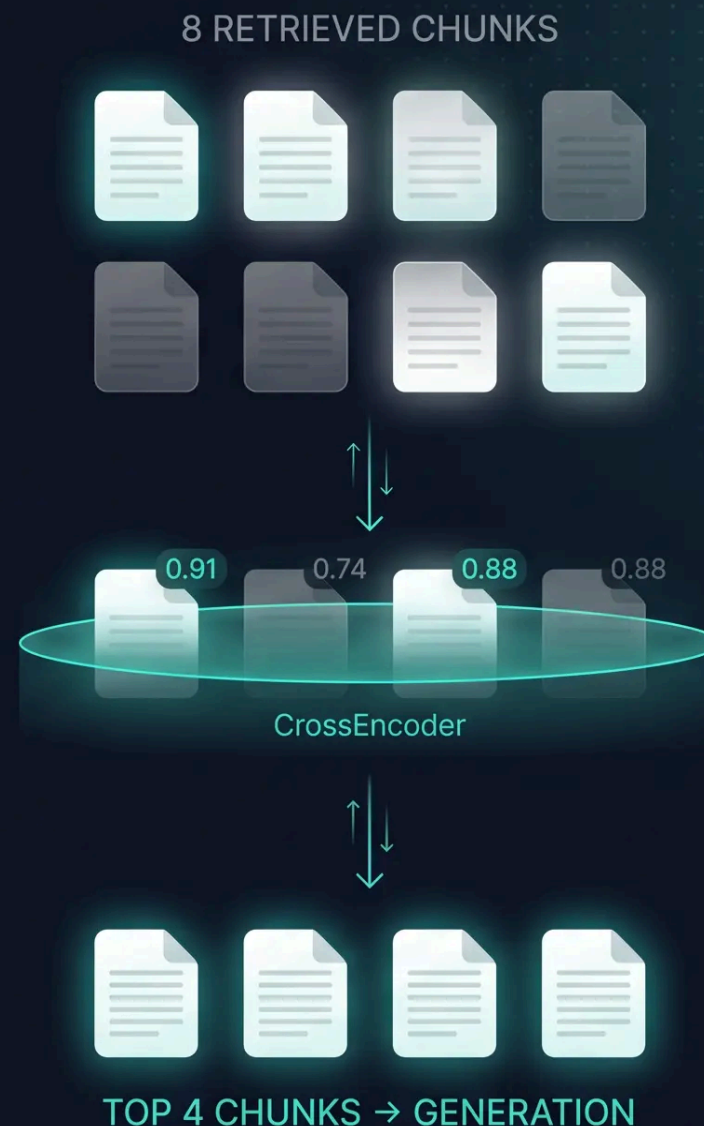
MEASURED OUTCOME

0.86

Faithfulness

0.87

Context Precision



Advanced Reasoning with LLM Agents

Block A — The Contrast

SIMPLE RAG

Handles direct, single-hop questions
One retrieval → One answer

LLM AGENT

Handles complex, multi-step reasoning
Compares, synthesizes, and cites across sections

Block B — Example Query Card

EXAMPLE QUERY

"Compare S1-14 health and safety requirements with general workforce disclosures"

3 tools used

2 reasoning steps

Structured output



Every Request. Fully Traced.

Nothing runs blind in production



PROMPT VISIBILITY

Full prompt sent to the LLM

Inspect every instruction, context window, and system message



RETRIEVAL SCORES

Chunks retrieved with relevance scores

Track which documents were fetched and how they ranked



LATENCY PER STEP

Time measured at each pipeline stage

Pinpoint bottlenecks across ingestion, retrieval, and generation



TOKEN CONSUMPTION

Input and output tokens logged per request

Monitor cost and efficiency across every query

“

You can't improve what you don't measure”

STEP 4 · EVALUATION

Performance Evaluation with RAGAS

12-question golden dataset · ESRS S1 corpus



FAITHFULNESS

0.86

Answer grounded in retrieved context



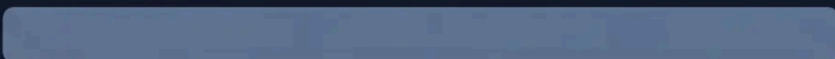
CONTEXT PRECISION

0.87

Relevant chunks ranked at the top

RETRIEVAL DEPTH OPTIMIZATION

k = 20 (baseline)



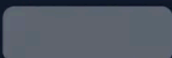
0.73

k = 8 (optimal) ✓



0.92

k = 4 (too few)



k = 8 selected as production default

KEY INSIGHT

“Measure, don’t guess — *k* matters more than expected”

Comprehensive Tech Stack

Every layer of the system, from API to deployment

API



FastAPI + Uvicorn

High-performance async REST API

EMBEDDINGS



Mistral Embed

Dense vector representations

RAG



LangChain · LangGraph

Retrieval pipelines & agent orchestration

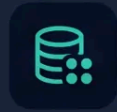
RERANKING



CrossEncoder ms-marco

Query-chunk relevance scoring

VECTOR STORE



Qdrant Cloud

Managed vector similarity search

LLM



Mistral Small

Cited answer generation

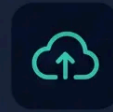
OBSERVABILITY



LangSmith

Full request tracing & evaluation

CLOUD · CI/CD



GCP Cloud Run · GitHub Actions

Containerized deployment & automation

DEPLOYMENT · LIVE

API Endpoints & Deployment

POST /chat/

RAG pipeline — hybrid retrieval + reranking

POST /agent/

LLM Agent — multi-step reasoning

POST /ingest/

Document ingestion into vector store

GET /health

Service health check

LIVE ON GCP CLOUD RUN

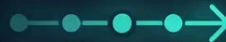
● esg-rag-chatbot-1057554534839.europe-west1.run.app

RAG Demo

Health & Safety Requirements — ESRS S1

USER QUERY

"What are the ESRS S1 requirements for employee health and safety?"



RAG RESPONSE

ESRS S1-14 requires entities to disclose material impacts on employee health and safety, including incident rates and management policies.

[1] · [2] · [3]

Source citations attached

Source: ESRS S1 — Own Workforce Standard

● Chunks Retrieved

8

● Chunks Used

4

● Citations Returned

3

Agent Demo: Comparative Analysis

Multi-step reasoning across regulatory sections

USER QUERY

"Compare S1-14 with general workforce disclosures"

TOOL CALL 1

compare_esrs_requirements
"S1-14" · "general workforce"

TOOL CALL 2

summarize_esrs_section
"ESRS S1 Table of Contents"

OUTPUT

Structured comparison table
with regulatory references

AGENT OUTPUT

Regulatory Comparison

Aspect	S1-14	General Workforce
Scope	--	--
Disclosures	--	--
Metrics	--	--

Sources: [ESRS S1, §14] · [ESRS S1, General]

Key Learnings & Insights

Four hard-won lessons from building a production RAG system



Hybrid Search Wins

Pure semantic search misses exact regulatory terms.
BM25 + semantic fusion is essential for regulatory corpora.



Measure Your k

The retrieval depth k directly impacts faithfulness.
Test it — k=8 outperformed k=20 in this corpus.



Define Your Tools Clearly

LLM agents are only as good as their tool docstrings.
Vague definitions lead to poor reasoning chains.



Observability Is Non-Negotiable

You can't debug what you can't see. LangSmith tracing revealed latency bottlenecks invisible without it.

"You can't improve what you don't measure."

THANK YOU · LET'S CONNECT

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github.com/Rudy-Nzau/esg-rag-chatbot

ESG RAG Chatbot · Production-grade LLM System



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Open to opportunities · AI & Data Science

Built for production. Grounded in sources. Ready for the real world.

●
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